

Program : Diploma in Printing Technology	
Course Code : 6101C	Course Title: Present Printing Industry
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To provide the basic idea on common printing practices and the related challenges existing in our printing industry.
- To provide an effective look up on the current printing industry practices and related challenges.
- To understand the importance and existence of ecofriendly printing and alternative technologies that may be adopted by the printing industry in the upcoming days.

Course Prerequisites:

Topic	Course Code	Course Title	Semester
Basics of various Printing process.		Basics of Printing processes.	2
Chemistry of wastes and its recovery.		Applied Chemistry	1
Wastes and Pollutions		Environmental Science.	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concept of present-day printing industry features and related output.	15	Understanding
CO2	Illustrate common press room practices and challenges carried out by conventional printing process.	14	Understanding
CO3	Explain about Industrial Waste characteristics, disposal and its control.	15	Understanding

CO4	Describe about the ways to reduce press room waste generation and to adopt eco friendly printing and alternative technologies.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2				2		
CO4	2				2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the concept of present-day printing industry features and related output.		
M1.01	Introduction to present printing industry.	4	Understanding
M1.02	Different printed products.	3	Understanding
M1.03	Common Printing Technologies.	4	Understanding
M1.04	Print market survey overview.	4	Understanding
Contents: Introduction to present printing industry. Different printed products - Commercial, Publication and Packaging products. Common Printing Technologies: Impact and Non-Impact Technologies, Importance of Conventional and Digital Printing. Print Market survey overview.			
CO2	Illustrate common press room practices and challenges carried out by conventional printing process.		
M2.01	Overview on present printing scenario in conventional printing.	3	Understanding

M2.02	Common Prepress practices.	3	Understanding
M2.03	Common Press practices.	3	Understanding
M2.04	Common Post-press practices.	2	Understanding
M2.05	Degree of automation in conventional printing process.	2	Understanding
M2.06	Challenges in - close register multicolor work.	1	Understanding
	Series Test I	1	

Contents:

Overview on present printing scenario in conventional printing process - Lithography, Flexography, Gravure, Screen and Digital printing. Common Prepress practices - Image generation, Image Processing, Proofs, Image carrier preparation. Press practices - Machine rudiments, Printing Process and Raw material basics. Post press practices - Machines involved and Common post press practices. Degree of automation in Lithography, Flexography, Gravure and Screen printing. Challenges in - close register multicolor work.

CO3	Explain about Industrial Waste characteristics, disposal and its control.		
M3.01	Industrial Waste characteristics in general.	3	Understanding
M3.02	Major Waste streams in Printing Press.	3	Understanding
M3.03	Pollution prevention control.	3	Understanding
M3.04	Common waste management practices.	3	Understanding
M3.05	Solvent Recovery process.	3	Understanding

Contents:

Industrial Waste characteristics in general - emission to air, discharge to water bodies, noise, and environmental hazardous waste. Major Waste streams in Printing Press. Pollution prevention control - Industry guidelines, Emission guidelines, Treatment Technologies. Common waste management practices -Waste Segregation, Disposal, Recycling, and Reuse. Solvent Recovery process.

CO4	Describe about the ways to reduce press room waste generation and to adopt ecofriendly printing and alternative technologies.		
M4.01	How to reduce waste generation at the beginning to end of a Press work.	5	Understanding
M4.02	Importance of eco friendly printing and alternative technologies.	5	Understanding
M4.03	Alternative Technologies used in printing industry.	4	Understanding
	Series Test II	1	

Contents:

How to reduce waste generation at the beginning to end of a Press work. Importance of eco-friendly printing and alternative technologies - Alternative Technologies used in Prepress, Press and Post press areas. Alternative Materials used for printing practices - alternative Inks and eco friendly substrates.

Text / Reference:

T/R	Book Title/Author
T1	Helmut Kipphan, Hand book of print media , Springer Science & Business Media.
R1	S.C. Sharma & M.P. Poonia, Environmental Studies , Khanna Publishing House, NewDelhi
R2	C.N.R.Rao Understanding Press Chemistry , Universities Press (India) Pvt. Ltd., 2011.
R3	N.R.Elred & T. Scarlet, Chemistry for the Graphic Arts , GATF, 1992.
R4	Tulika Das, Chemistry in Printing, 2nd Edition , Barnana Prakashani, 2011.
R5	Aldo Vieira, Da Rosa, Fundamentals of renewable energy processes , Academic Press Oxford.

Online Resources:

Sl.No	Website Link
1	https://en.wikipedia.org/wiki/Printing_industry_in_India
2	https://en.wikipedia.org/wiki/Waste_management
3	https://www.unescap.org/sites/default/files/CH08.PDF
4	http://printwiki.org/Printing
5	https://en.wikipedia.org/wiki/Printing